

CS & IT ENGINEERING

Algorithm

Miscellaneous

Lecture No. 1



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Topic

Heaps

Topics to be Covered



Topic

Topic

Topic

Misc : TC analysis using
Recursion Tree.



About Aditya Jain sir



1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.





Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

Sat 2 Aug → 11:30 - 1:30

Sun 3 Aug → 1) 8:30 - 10:30

2) 12:30 - 2:30 (last lec)



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Time Complexity Analysis : (Recurrence Relations)

Approaches	We can find
Back-substitution	Value of Recurrence TC: notation
Master Method / Theorem	TC: notation
Recurrence / Recursion Tree Approach	TC: notation



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Types of TC Recurrences :

$D_n C$

1. Symmetric : $T(n) = a \times T(n/b) + f(n), a \geq 1, b > 1, f(n) = +ve$

2. Asymmetric

mm

(a) Case 1 : $T(n) = T(\alpha n) + T((1-\alpha)n) + f(n)$

(b) Case 2 : $T(n) = T(\alpha n) + T(\beta n) + T(\delta n) \dots + f(n)$



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Symmetric Recurrence:

$$T(n) = a \times T(n/b) + f(n), a \geq 1, b > 1, f(n) = +ve$$



Number of sub-problem

Size of each sub-problem

Can be solved by:

1. Back substitution ✓
2. Master method ✓
3. Recursion tree approaches

Equation:

$$T(n) = \cancel{2T(n/2)} + \cancel{2T(n)} = 2T(n/2) + n$$



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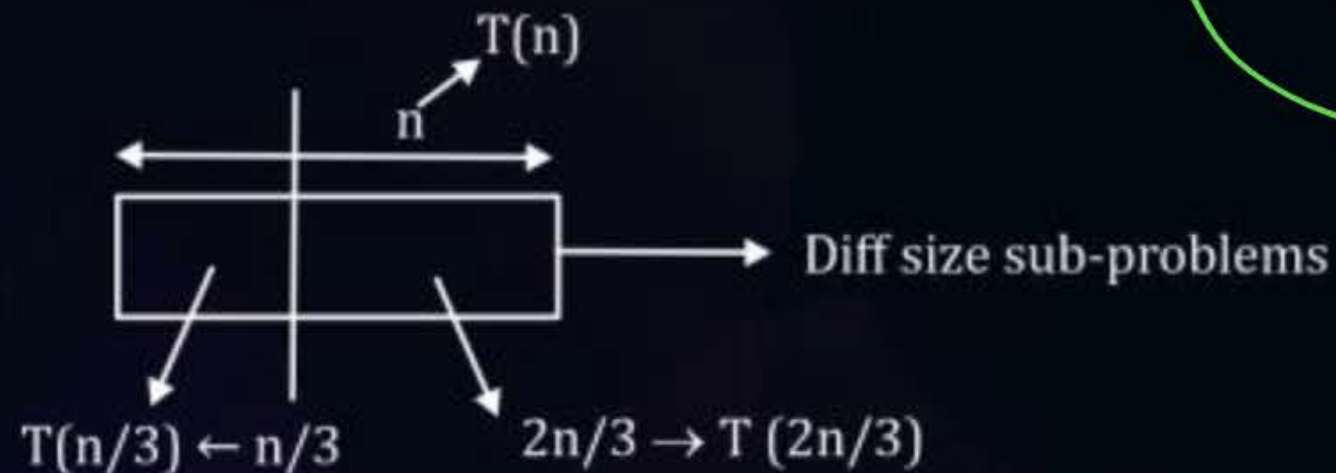
Asymmetric:

Case-1

$$T(n) = T(\alpha n) + T((1-\alpha)n) + f(n)$$

$$T(n) \quad 0 \leq \alpha \leq 1$$

Equation:



Rec Tree Tc

$$1. \quad T(n) = T(n/3) + T(2n/3) + n$$

$$2. \quad T(n) = T(n/4) + T(3n/4) + n^2$$

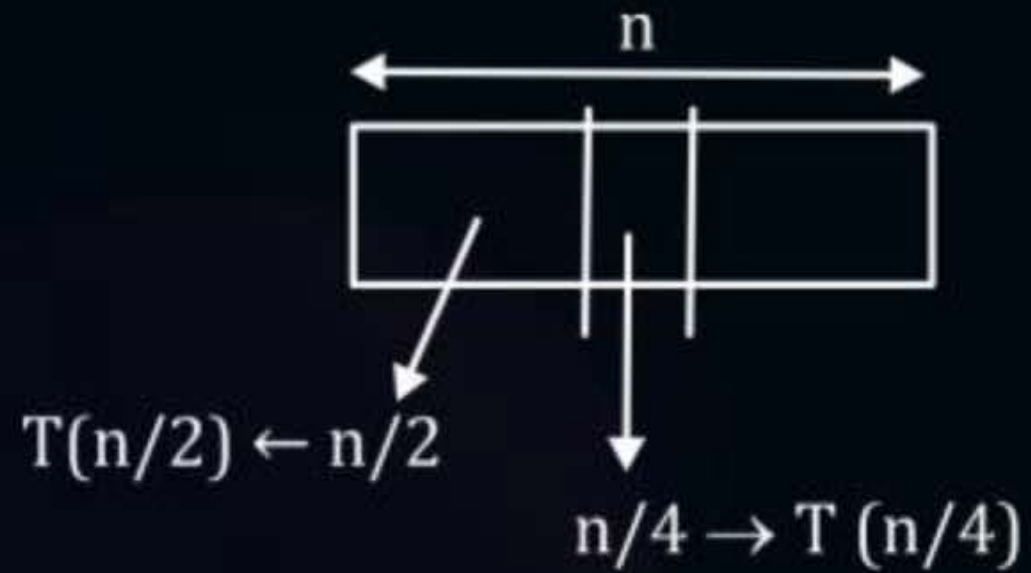


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Asymmetric:

Case-2



$$T(n) = T(n/2) + T(n/4) + n^2$$

Recursion tree approach.



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Equation:

$$T(n) = 2T(n/2) + n \longrightarrow \text{Merge sort}$$

1. ✓ Back $\rightarrow$ substitution $\rightarrow$ value of recurrence?
2. Master method.



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Solution:

Back- substitution:

$$T(n) = 2T(n/2) + n \text{_____} (1)$$

$$T(n/2) = 2T(n/2^2) + n/2$$

$$T(n) = 2T(n/2^2) + n/2 + n$$

$$T(n) = 2^2 T(n/2^2) + 2n \text{_____} (2)$$

$$T(n) = 2^3 T(n/2^3) + 3n \text{_____} (3)$$



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General Term

$$T(n) = 2^k T(n/2^k) + k * n \quad (4)$$

$$\text{for B-C, } \frac{n}{2^k} = 1 \Rightarrow n = 2^k$$

$$k = \log_2^n$$

$$T(n) = D * T(1) + n * k$$

$$T(n) = n * c + n \log_2^n \rightarrow \text{value of recurrence}$$

$$T(n) = O(\log_2^n)$$

$$O(n \log_2^n)$$



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Master Method:

$$T(n) = 2T(n/2) + n \quad (\text{given})$$

$$T(n) = a * T(n/b) + f(n)$$

$$a = 2$$

$$b = 2$$

$$\cancel{f(n) = 2}$$

$$f(n) = n$$

Valid

$$\log_b^a = \log_2^2 = 1$$



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Case-1:

Is $f(n) = \theta(n^{\log \cdot \epsilon})$ $O(n^{1-\epsilon})$

Is $n = \theta(n^{1-\epsilon})$, Some $\epsilon > 0$? $\times$ Invalid

Case-2:

Is $F(n) = \theta(n^{\log_b^a} * (\log n)^k)$, for some

(a) $k \geq 0$

For $k = 0$, $n = \theta(n^1 * (\log n)^0)$

$n = \theta(n)$

Valid to for $k = 0$



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Then,

$$T(n) = \theta(n \log_b^1 * (\log n)^{k+1})$$

$$k = 0 \Rightarrow T(n) = \theta(n^1 * (\log n)^{k+1})$$

$$T(n) = \theta(n \log n)$$



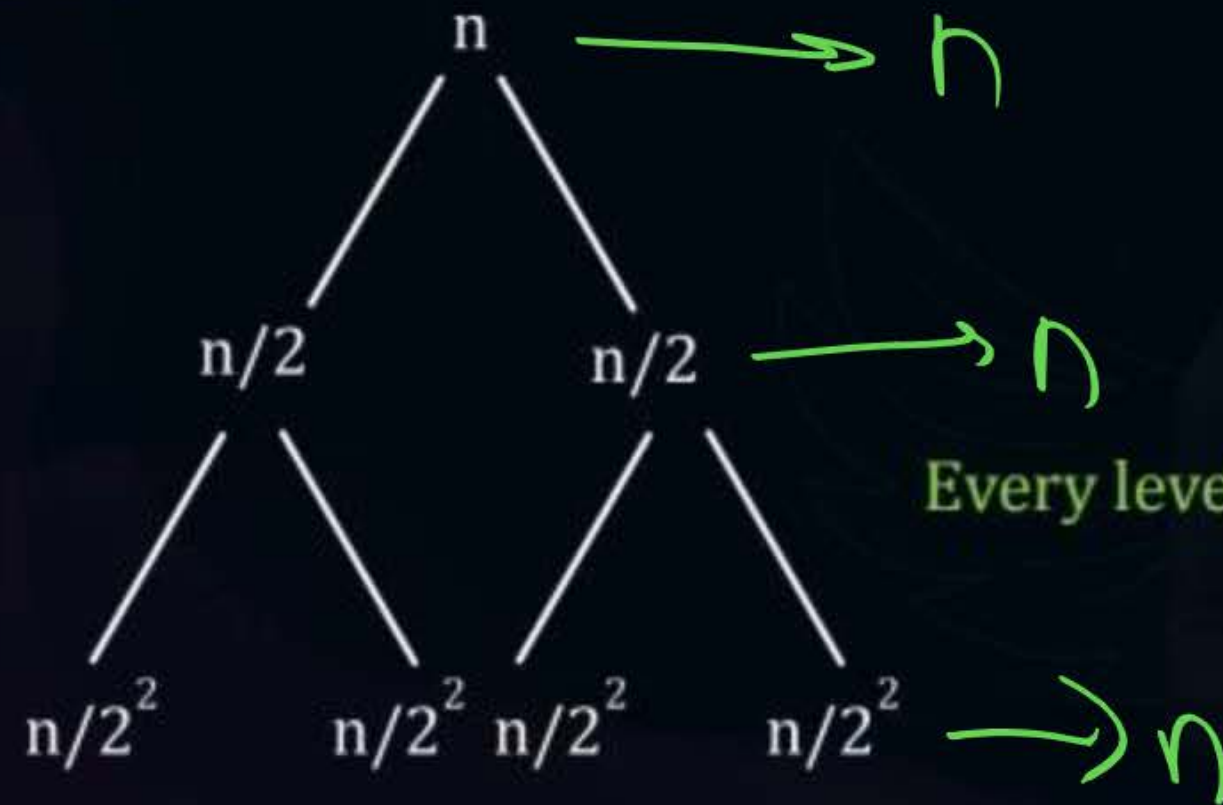
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Recurrence tree Approach:

$$T(n) = 2T(n/2) + n$$

↓
Cost/fee of recursion



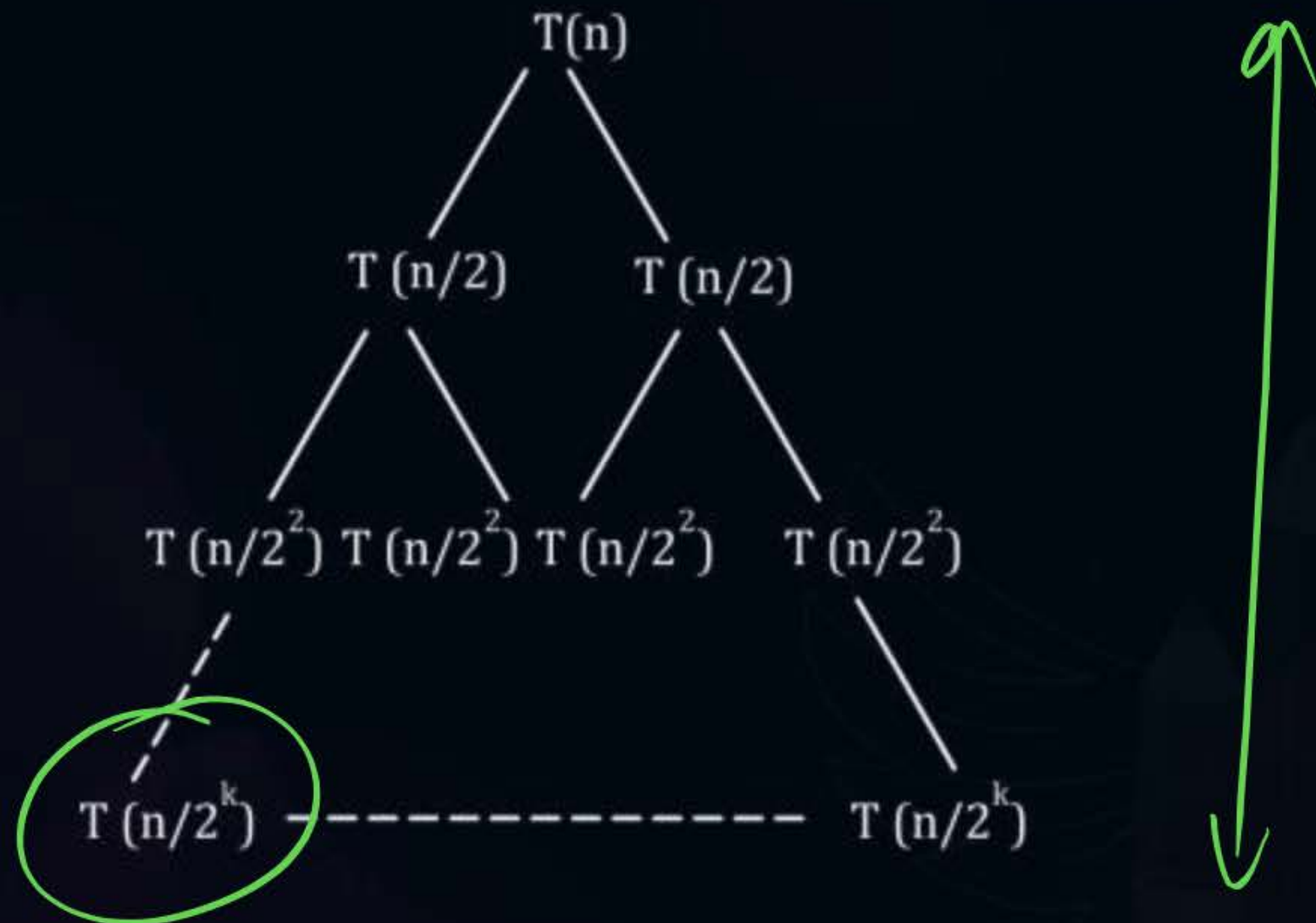
Every level has cost of 'n'.



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K levels



Assume this is last level (Base condition)



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At last level, it should be base condition

$$T(n/2^k)$$

$$\boxed{\frac{n}{2^k} = 1}$$

$$n = 2^k$$

$$\boxed{k = \log_2 n}$$

$$K = \log_2 n$$



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Level -1 $\rightarrow T(n)$: cost = n

Level -2 $\rightarrow T(n/2)$: cost = n

Level -3 $\rightarrow T(n/2^2)$: cost = n

Level $-(k+1) \rightarrow T(n/2^k)$: cost = n



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Total number of levels = $(k+1)$

The cost/ time of every level = n

Hence, total cost/ time = $n * (k+1)$
 $= n * k + n$

We know

$$k = \log_2^n$$

$$= n * \log_2^n + n$$

$$T(n) = \theta(n \log n)$$

$$T(n) = T(n/3) + T(2n/3) + n$$



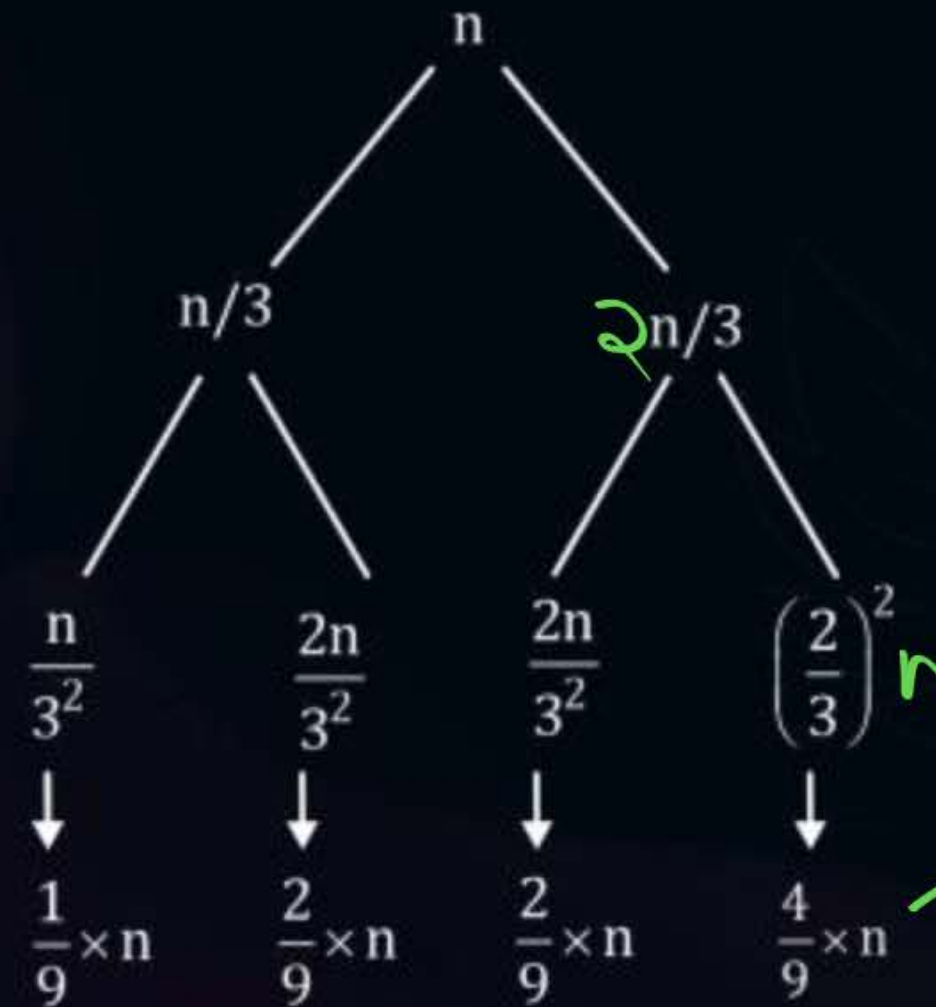
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Equation:

$$T(n) = T\left(\frac{n}{3}\right) + T\left(\frac{2n}{3}\right) + n$$

Asymmetric case -1



Cost at every level = n

$$n \rightarrow \left(\frac{1}{9} + \frac{2}{9} + \frac{2}{9} + \frac{4}{9} \right) n$$

$$= \frac{1+2+2+4}{9} \times n$$

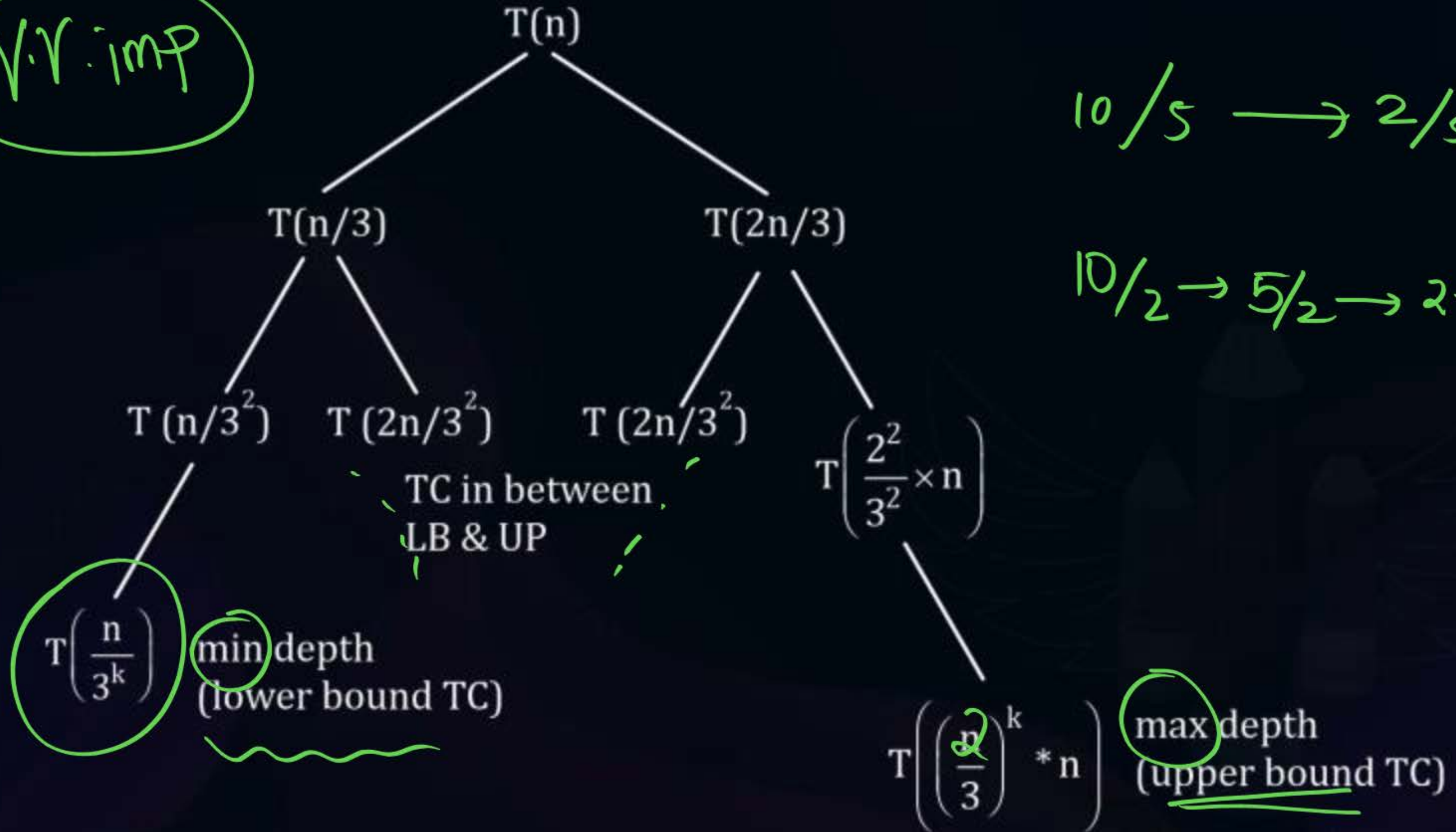
$$= \frac{9}{9} \times n$$

$$= 1 \times n$$



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V.V.IMP

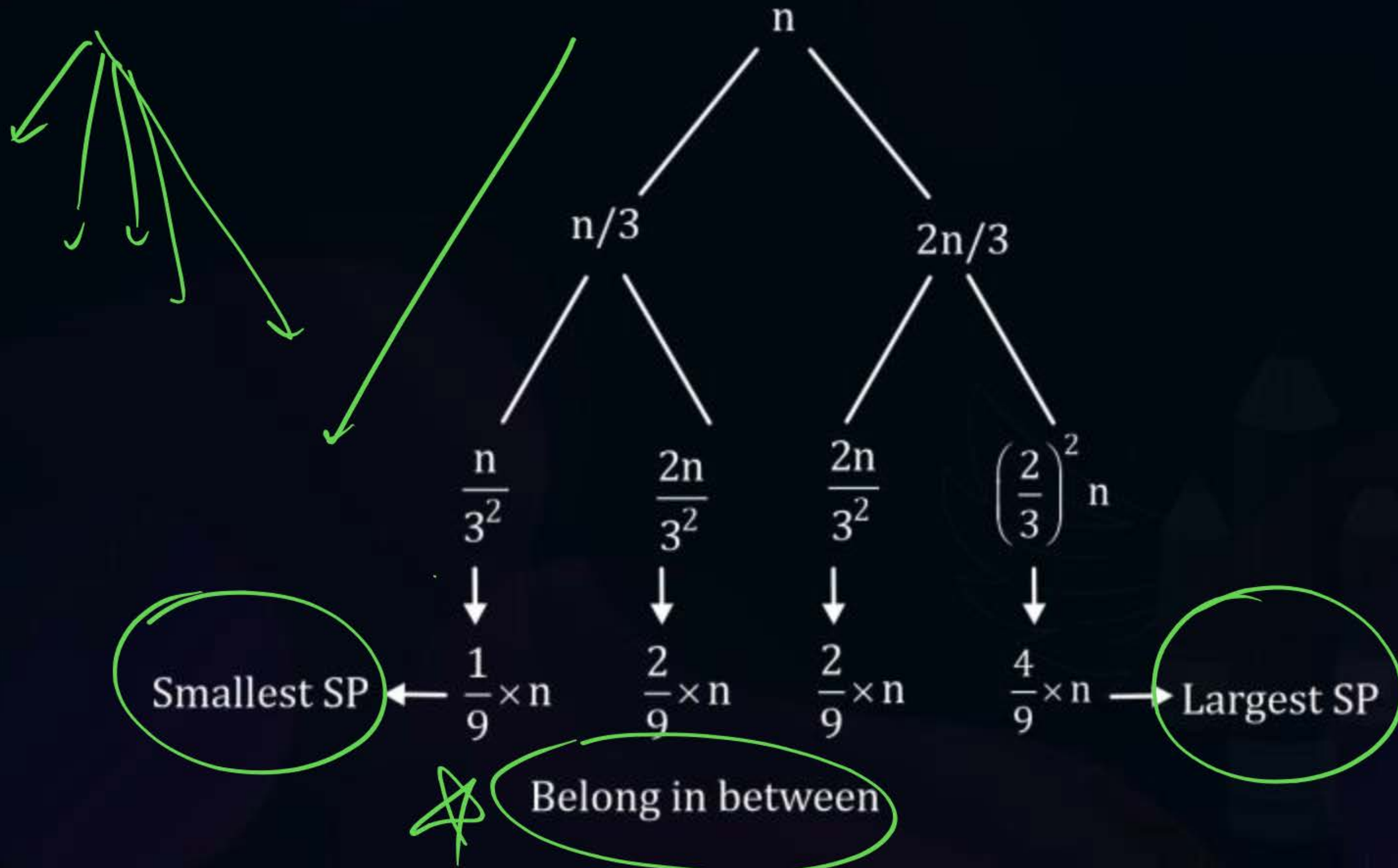


$$10/5 \rightarrow 2/5 \rightarrow 0$$

$$10/2 \rightarrow 5/2 \rightarrow 2.5/2 \rightarrow 1.25 \dots$$

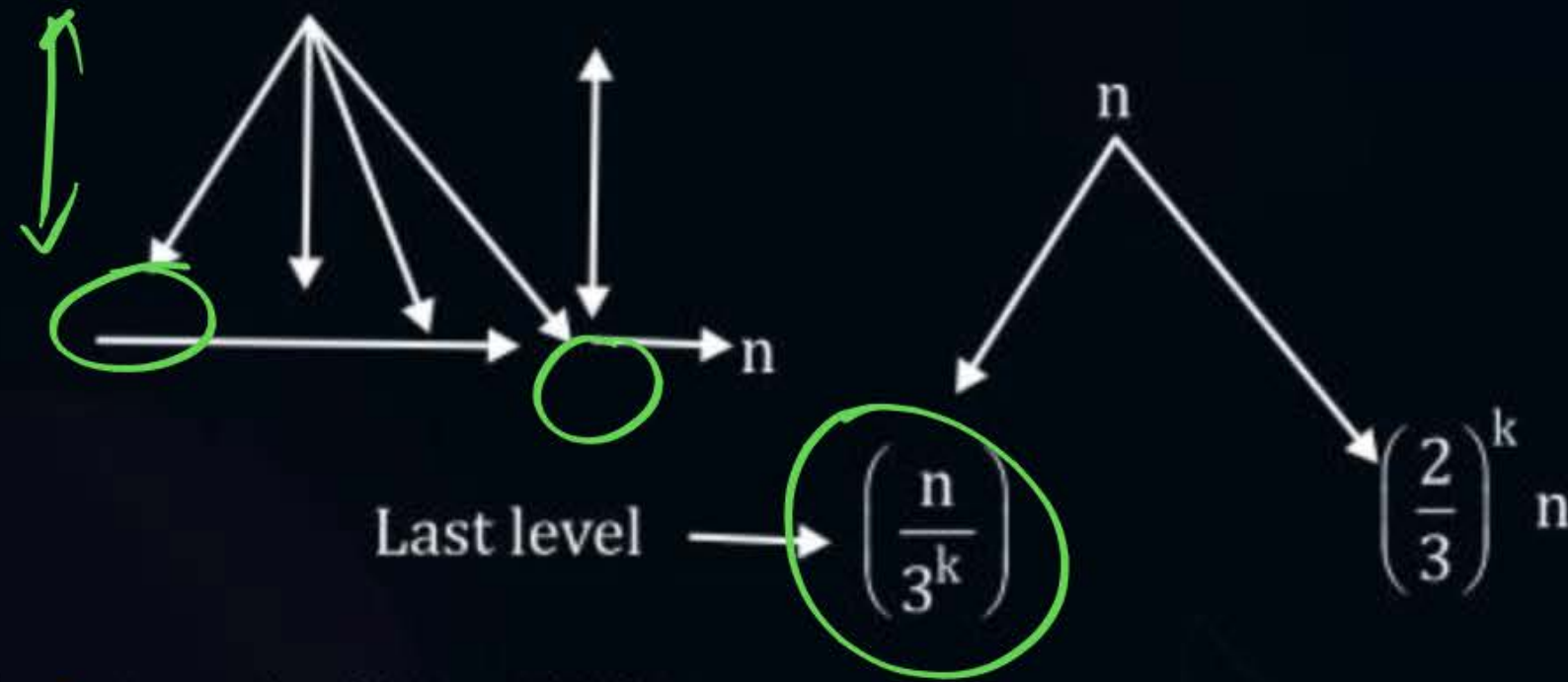


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1. For smallest/ min depth (LB of TC)

At least level $k, \frac{n}{3^k} = 1$ (BC)

$$n = 3^k$$

$$k = \log_3 n$$



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Assuming there are k levels and cost/time at each level = n

Hence, $T(n) \geq k * n$

$$T(n) \geq \log_3^n * n$$

$$T(n) \geq n * \log_3^n \quad \text{--- (i)}$$

$$T(n) = \Omega(\log_3^n * n)$$



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For largest depth $\rightarrow$ (UB of TC)

Let last level be k

For last level $\left(\frac{2}{3}\right)^k * n = 1$

level ~~$\left(\frac{2}{3}\right)^k = n$~~ $\left(\frac{3}{2}\right)^k = n$

$$\log n = k * \log \left(\frac{3}{2}\right)$$

$$\frac{\log^a}{\log^b} = \log_b^a$$

$$k = \frac{\log n}{\log \left(\frac{3}{2}\right)} = \log_{3/2}^n$$

$$k = \log_{3/2}^n$$



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Cost at every level = n

Total cost $\leq k * n$

$$T(n) \leq \cancel{n} \log_{3/2}^n * n$$

$$T(n) \leq n * \log_{3/2}^n \text{----- (ii)}$$

~~$$T(n) = O(\log_{3/2}^n)$$~~

$$T(n) = O(n \log_{3/2} n)$$



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From (i) & (ii)

$$n * \log_3^n \leq T(n) \leq n * \log_{3/2}^n$$

$$T(n) = \Omega(\log_3^n * n)$$

$$T(n) = \Omega(n * \log_{3/2}^n)$$

Approximation $\Rightarrow T(n) = \theta(n \log n)$

~~INATE~~



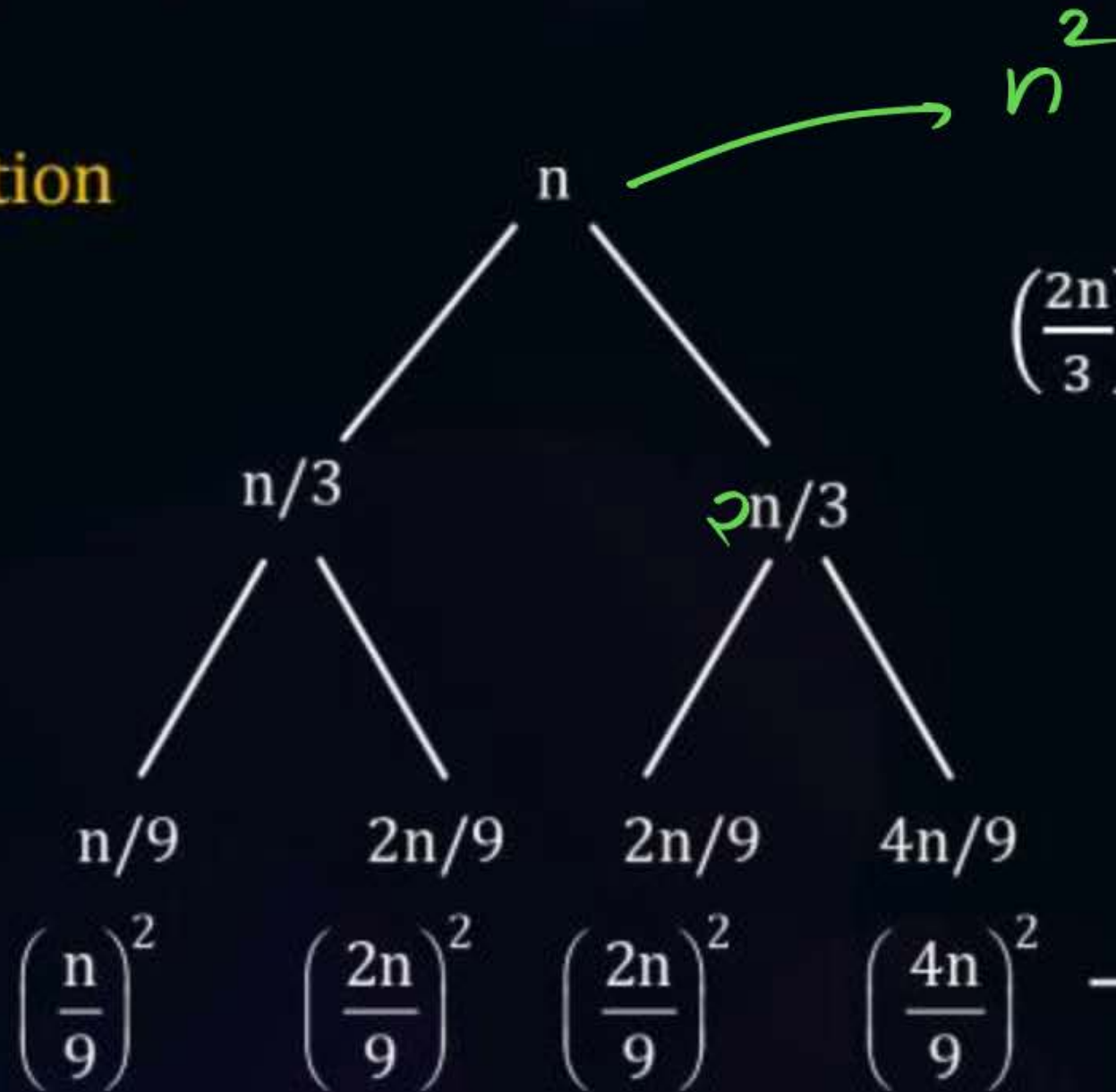
#Q. $T(n) = T\left(\frac{n}{3}\right) + T\left(\frac{2n}{3}\right) + n^2$



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Solution



$$\left(\frac{2n}{3}\right)^2 + \left(\frac{n}{3}\right)^2 + \cancel{\left(\frac{2n}{3}\right)^2} + \cancel{\frac{n^2}{9}} + \cancel{\frac{4n^2}{9}} = \frac{5n^2}{9}$$

$$\frac{n^2}{9^2} + \frac{4n^2}{9^2} + \frac{4n^2}{9^2} + \frac{16n^2}{9^2} = \frac{5n^2}{9^2}$$

$$\longrightarrow \left(\frac{5}{9}\right)^2 n^2$$



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Let there be k levels

Total at each level

1 st Level	→ n^2	→ $\left(\frac{5}{9}\right)^0 n^2$
2 nd Level	→ $\frac{5}{9}n^2$	→ $\left(\frac{5}{9}\right)^1 n^2$
3 rd Level	→	$\left(\frac{5}{9}\right)^2 n^2$
k th Level	→	$\left(\frac{5}{9}\right)^{k-1} n^2$





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Hence, Total cost (T(n))

⇒ Sum of cost at each level.

$$T(n) = \left(\frac{5}{9}\right)^0 n^2 + \left(\frac{5}{9}\right)^1 n^2 + \dots + \left(\frac{5}{9}\right)^{k-1} n^2$$

$$= \sum_{i=0}^{k-1} \left(\frac{5}{9}\right)^i n^2$$

$$= \frac{1}{\left(1 - \frac{5}{9}\right)} * n^2$$

$$\text{Total cost} \approx \frac{9}{4} n^2$$

$$\rightarrow O(n^2)$$



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Shortest

$$\sum_{i=0}^{\infty} x^i = \frac{1}{1-x}$$

(approximation)



$$T(n) = \theta(n^2)$$

K* cost all each level

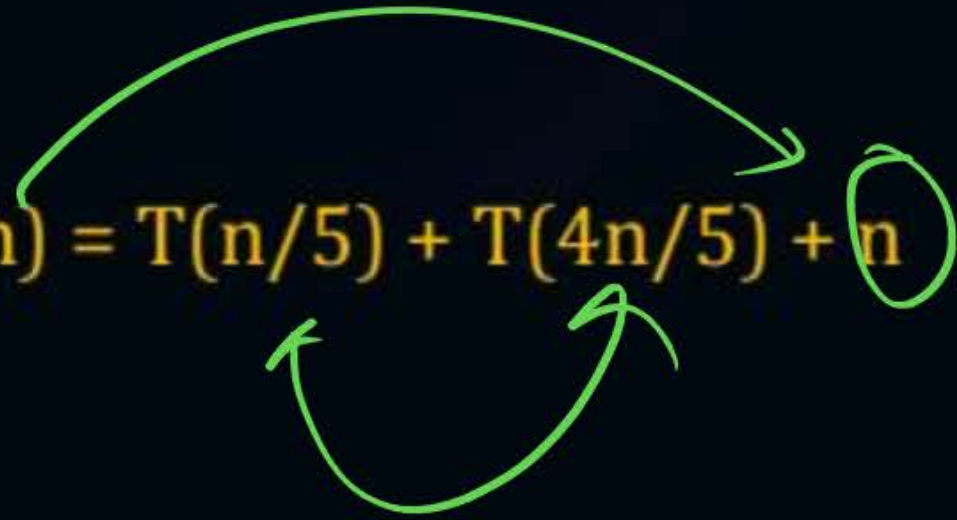
here it is difference

Hence, no.

[NAT]

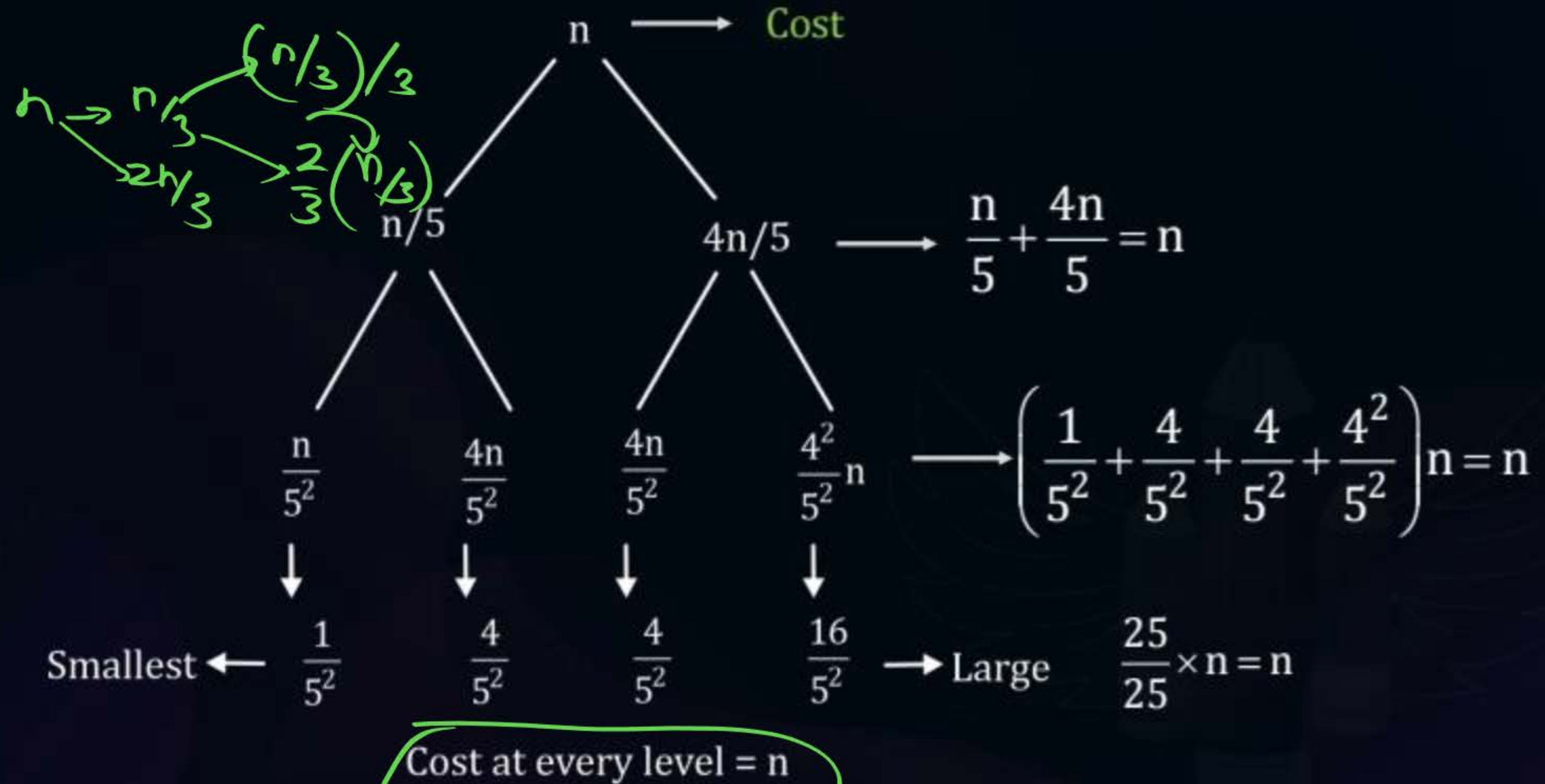


#Q. $T(n) = T(n/5) + T(4n/5) + n$



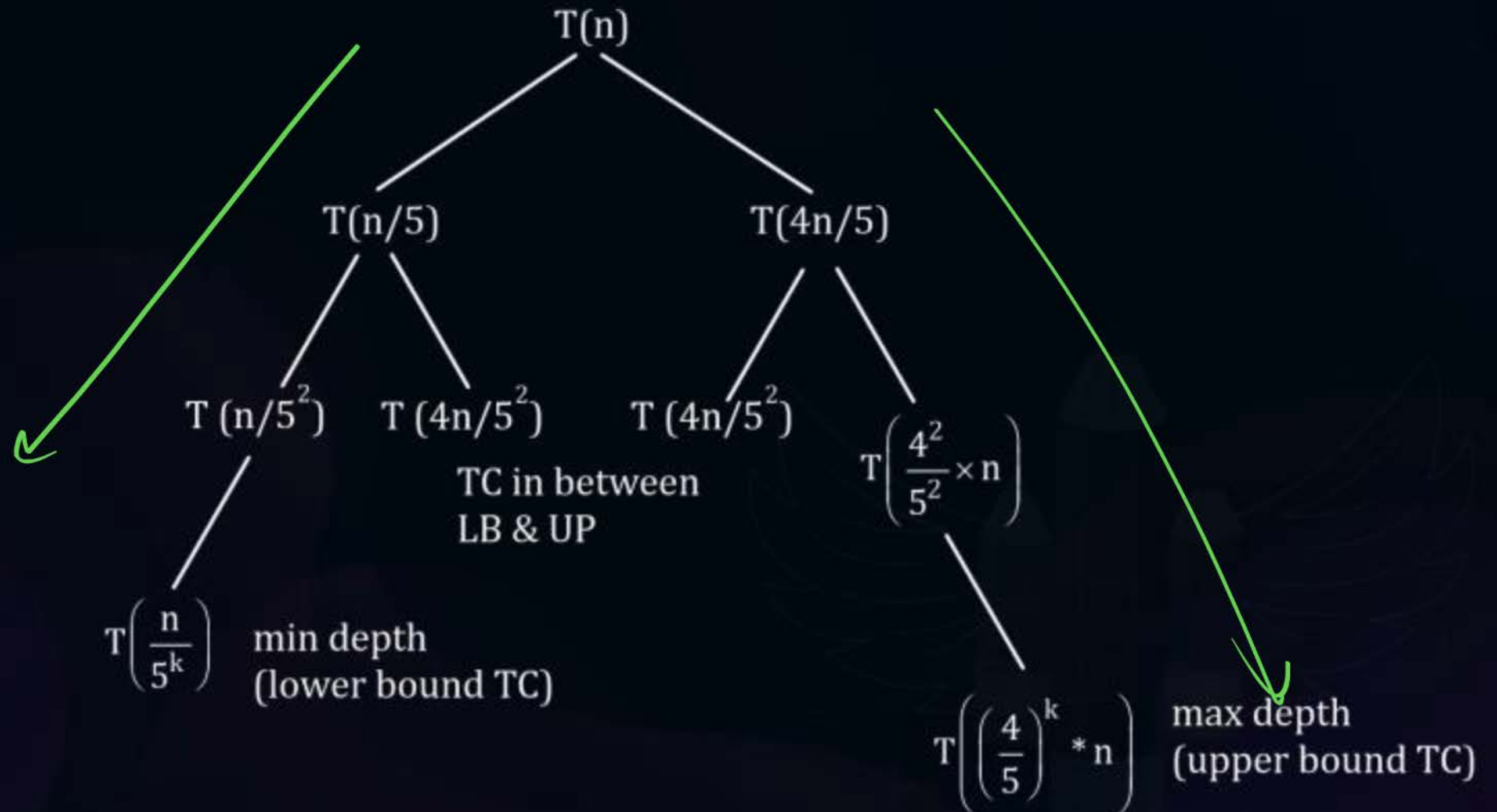


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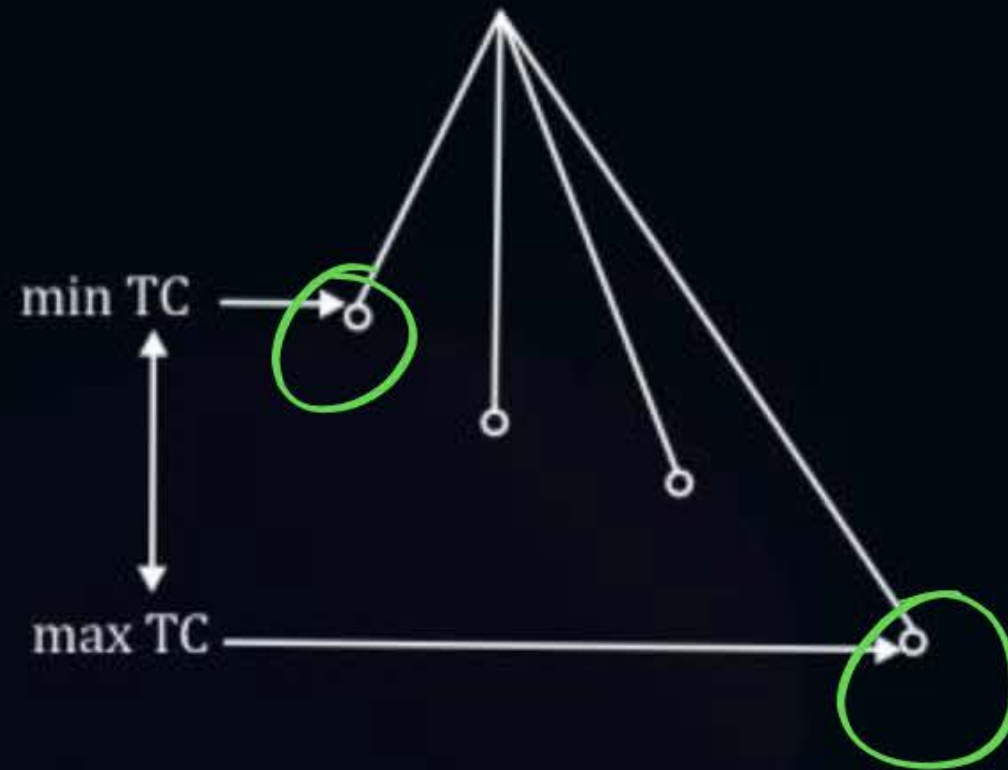


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For

Lowest depth (LB) TC

$$\frac{n}{5^k} = 1$$

$$5^k = n$$

$$k = \log_5^n$$

Cost at only level = n

$$T(n) \geq n * k$$

$$T(n) \geq n * \log_5^n$$

$$T(n) = \Omega(n \log_5^n)$$





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For largest depth:

$$\left(\frac{4}{5}\right)^k n = 1$$

$$\left(\frac{5}{4}\right)^k = n$$

$$k = \log_{5/4}^n$$

$$T(n) < n * k$$

$$T(n) < n * \log_{5/4}^n$$





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$$n * \log_5^n \leq T(n) \leq n * \log_{5/4}^n$$

$$T(n) = \Omega(n \log_5^n)$$

$$T(n) = \Omega(n \log_{5/4}^n)$$

Approximation: $T(n) = \theta(n \log n)$

$T(1)=1$

$T(n) = T(n-1) + T(n-2) + 1$

Approx: $T(n-1) > T(n-2)$
 $T(n-2) \approx T(n-1)$

$T(n) = T(n-1) + T(n-1) + 1$

$T(n) = 2T(n-1) + 1$

② Recursion Tree

HW

Back-sub



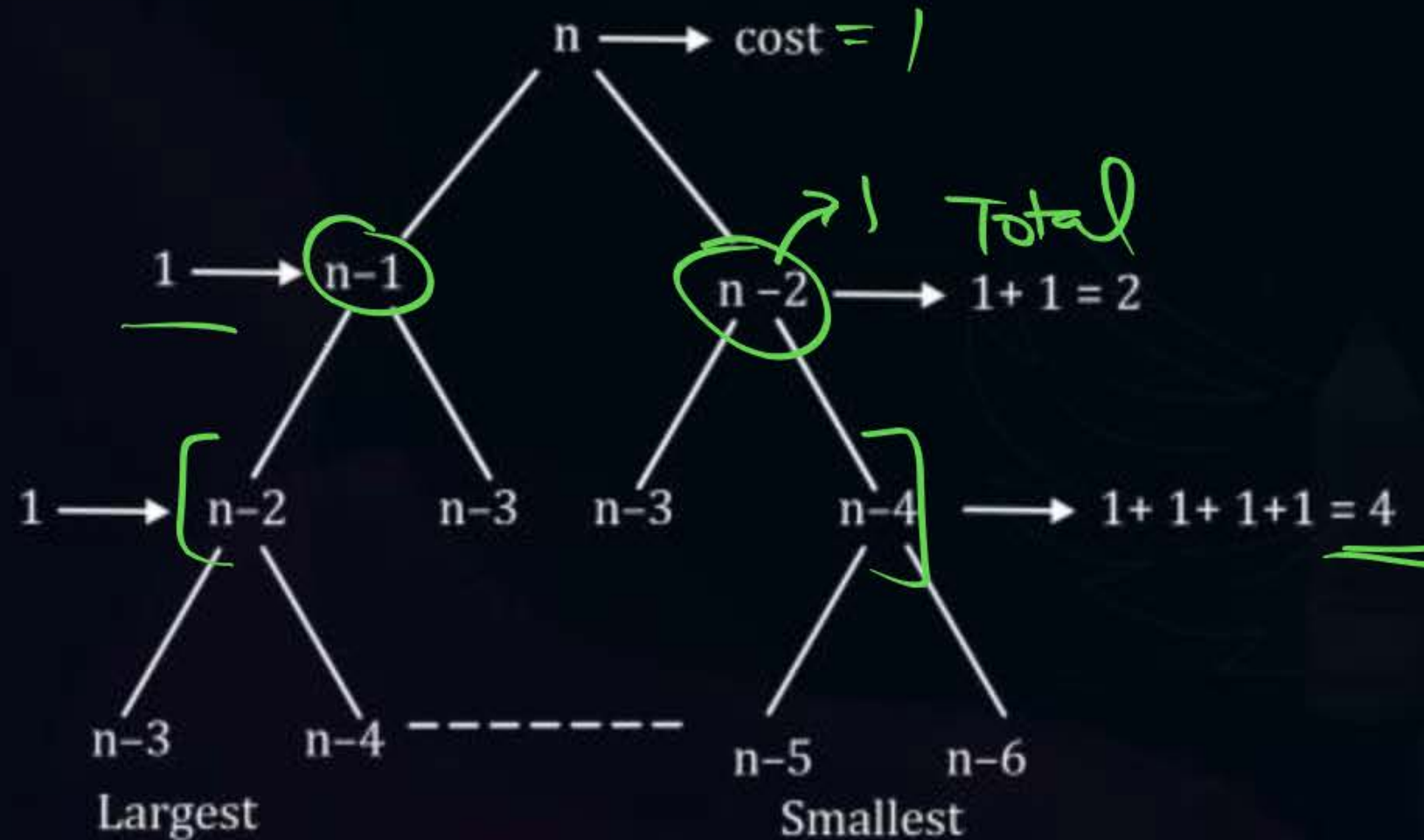
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Fibonacci:

TC ~~rearrange~~

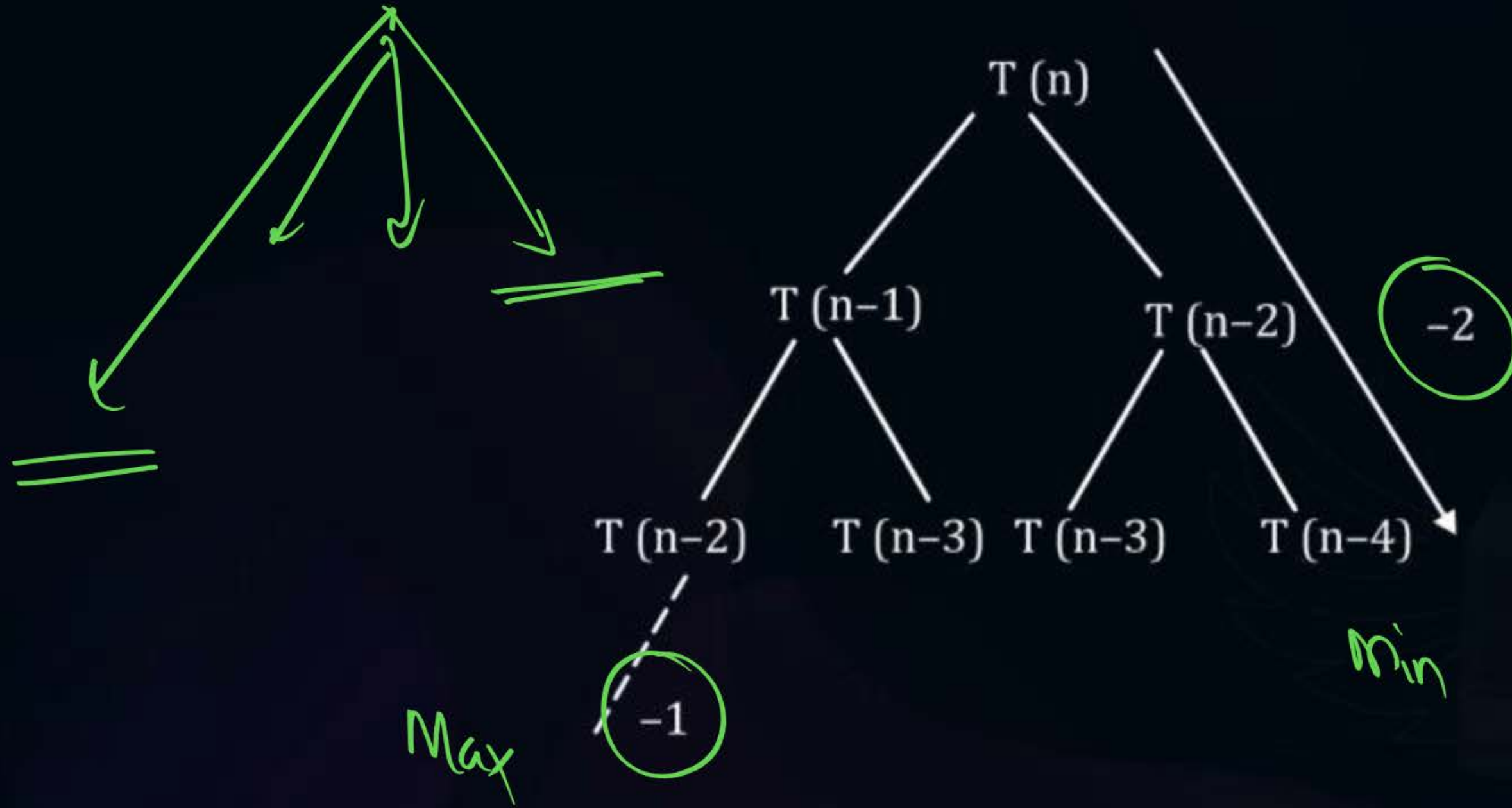
$$T(n) = T(n-1) + T(n-2) + 1$$

Recurrence





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Left -most branch

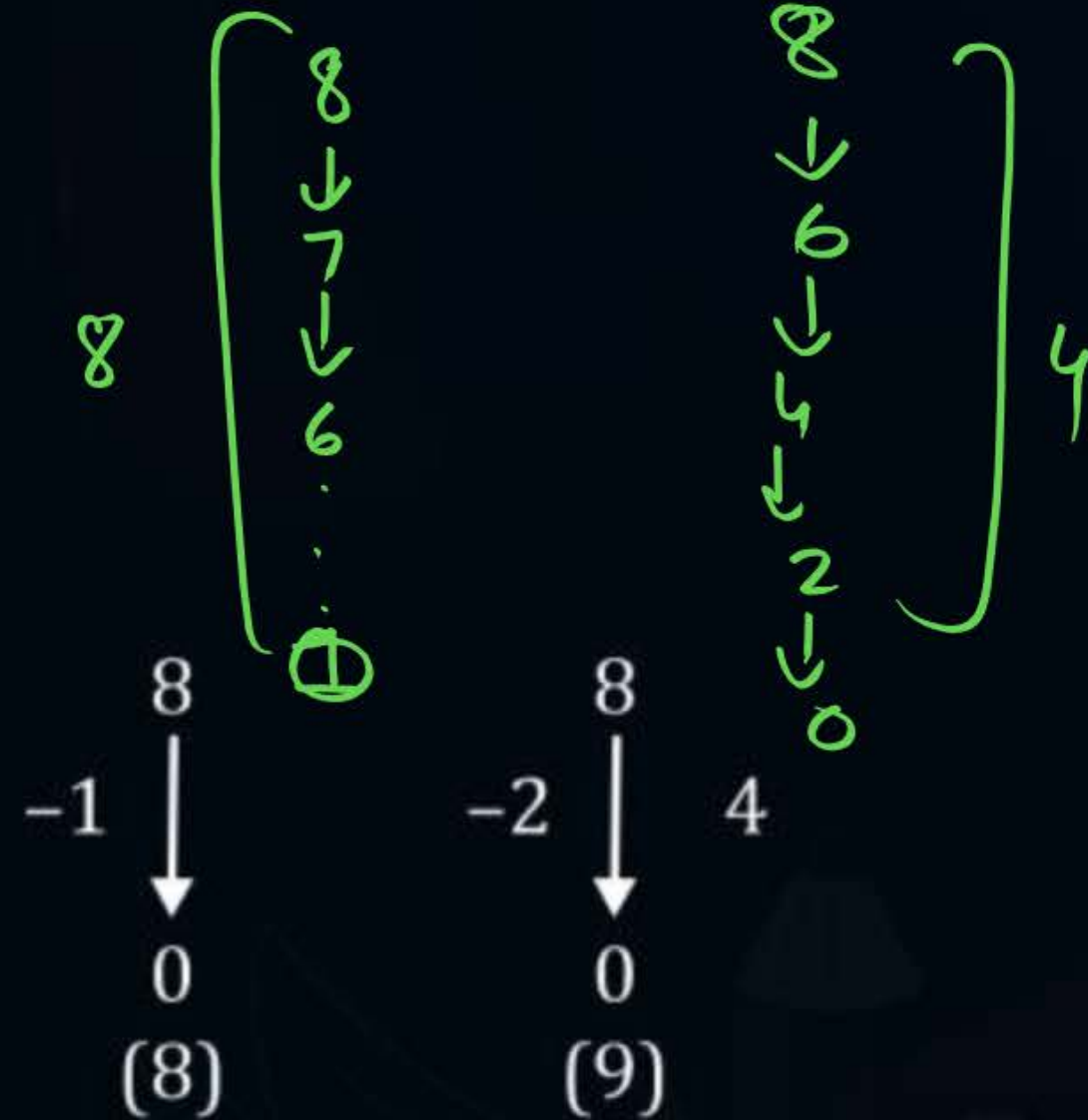
$n-1 \rightarrow n-2 \rightarrow n-3 \dots\dots\dots$

Number of levels $K = n$ (UB TC)

Right -most branch

$n-2 \rightarrow n-4 \rightarrow n-6 \dots\dots\dots$

Number of levels = $n/2$ (LB of TC)





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Total cost of k levels $= 2^0 + 2^1 + \dots + 2^{k-1}$

~~Cost of answer level k~~

$$= \cancel{2^0 + 2^1 + 2^2 + \dots + 2^{k-1}}$$

$$= \cancel{(2^k - 1)}$$

$$\sum_{i=0}^{k-1} 2^i$$

$$= 2^k - 1$$

1. Left most branch $\rightarrow k = n$

$$T(n) \leq 2^n - 1$$

$$T(n) = O(2^n)$$



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2. Right most branch $\rightarrow k = n/2$

$$T(n) < 2^{n/2} - 1$$

$$T(n) = \Omega(2^{n/2})$$



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Conclusion:

Fibonacci

$$T(n) = O(2^n)$$

$$\boxed{\cancel{T(n) = \Omega(n/2^n)}}$$

$$T(n) = \Omega(2^{n/2})$$

It is $\theta(2^n)$?

No



2 mins Summary



TC using Rec Tree



THANK - YOU